# Slat control and monitoring system

Description of function

| | | | |
|---|---|---|---|
| **Object code** | DMC-MRDX-A-27-81-00-00A-042A-A | **Issue** | 003-00 (revised) |
| **Issue date** | 2026-03-12 | **Language** | en-GB |
| **Responsible partner** | AERALIS S.A.S. (U8025) | **Originator** | AERALIS S.A.S. — Technical Data Valmont (U8025) |
| **Security** | 02 — Restricted | **Quality assurance** | First verification (tabtop) |
| **Applicability** | MRDX-42 and MRDX-46, all aircraft | | |

## References

| CODE | TITLE |
|---|---|
| DMC-MRDX-A-27-81-00-00A-720A-A | Slat actuation power control unit — Installation |
| DMC-MRDX-A-27-80-00-00A-040A-A | High lift system — Description |

## 1 General

The slat control and monitoring system moves the leading edge slats between the retracted position and the four extended positions selected on the flap and slat control lever. The system is made of two slat flap control computers (SFCC), one power control unit (PCU) with two hydraulic motors, a torque shaft transmission along the leading edge of each wing, and the wing tip brakes that hold the transmission if an asymmetry is detected.

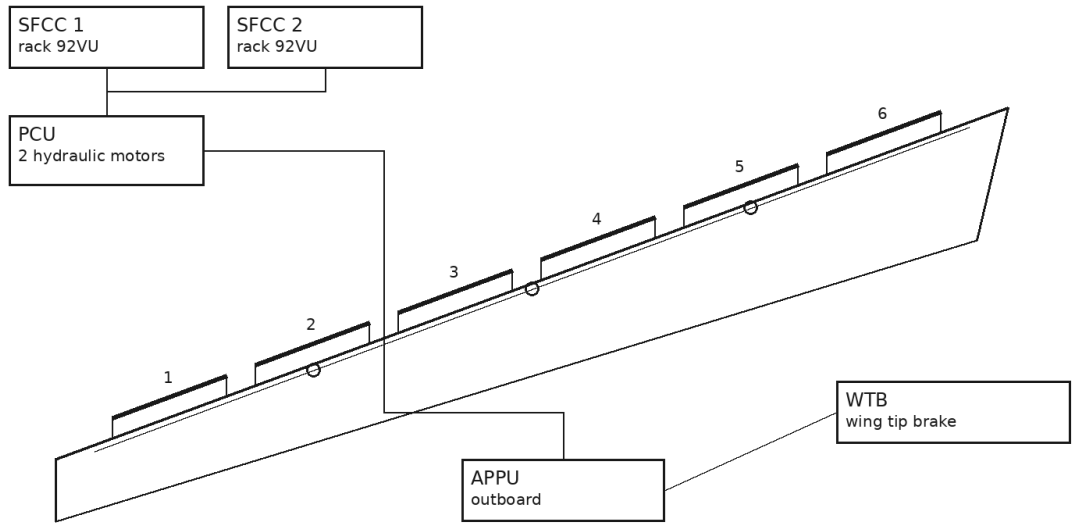
Each SFCC has one slat channel and one flap channel. The two computers work in an active/standby arrangement: the master computer commands the PCU while the standby computer monitors the command and the position feedback. If the master computer fails, control passes to the standby computer without an interruption of the surface movement.

> The position feedback comes from two feedback position pick-off units (FPPU) installed on the PCU output and from one asymmetry position pick-off unit (APPU) installed at the outboard end of each transmission. The APPU signals are compared continuously; a difference of more than 3.6 degrees of transmission rotation between the left and the right wing causes the wing tip brakes to engage **and latches the slat system** until the fault is cleared on the ground.

## 2 Component location

The PCU is installed in the belly fairing between frame C41 and frame C44, on the left hand side of the centre line. Access is through the fwd belly fairing access panel 196AB. The two SFCC are installed in the avionics compartment, in rack 92VU.

SLAT CONTROL AND MONITORING SYSTEM — COMPONENT LOCATION (LH WING)




1 to 6 slat panels SFCC slat flap control computer PCU power control unit

**Slat control and monitoring system — component location — Figure 1**

**NOTE:** The APPU on the right hand wing is accessible only when the slats are fully retracted and the hydraulic systems are depressurized.

## 3 Operation

### 3.1 Normal operation

When the flight crew moves the control lever, the two SFCC read the lever position from two independent position sensors. The master computer calculates the commanded slat position, compares it with the FPPU feedback and energizes the PCU valve block until the difference is inside the tolerance band given in Table 1.

**Table 1 Slat position tolerance band**

| Lever position | Slat angle (deg) | Tolerance (deg) | Placard speed (kt) |
|---|---|---|---|
| 0 | 0.0 | ±0.5 | — |
| 1 | 14.5 | ±0.8 | 263 |
| 2 | 19.0 | ±0.8 | 222 |
| 3 | 23.0 | ±1.0 | 196 |
| FULL | 27.0 | ±1.0 | 180 |

### 3.2 Alternate operation

**APPLICABLE TO: Aircraft post-modification 27-8104 embodied**

After modification 27-8104, a slat runaway that is detected during the extension sequence no longer causes both hydraulic motors to be isolated. The affected motor is isolated and the remaining motor drives the transmission at half rate. The flight crew gets the `SLATS SLOW` memo on the engine and warning display.

**CAUTION**

Do not dispatch the aircraft with one slat motor isolated unless the dispatch conditions of the minimum equipment list are satisfied. The landing distance penalty is not applied automatically by the flight management system.

## 4 Interfaces

- Hydraulic: green and yellow systems, through the PCU valve block.
- Electrical: 28 V DC essential bus for SFCC 1 and 28 V DC bus 2 for SFCC 2.
- Avionics: two AFDX ports per computer to the flight control and guidance function.